

Authentication Overhead in Post-Quantum IoT for Classical, Full-PQC and Hybrid TLS

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Abstract—The transition to post-quantum cryptography is particularly challenging for resource-constrained IoT devices, where improved cryptographic resilience must be balanced against latency, energy, and reliability constraints. This paper presents an empirical evaluation of TLS 1.3 authentication overhead on ESP32 under three key-exchange modes: classical X25519, full post-quantum ML-KEM-768, and hybrid X25519+ML-KEM-768. The study contributes the first direct large-scale comparison of these regimes on Xtensa-class hardware using wolfSSL and introduces IoT-QARS, an extension of the PAREK risk model that incorporates measured deployment overhead. Across 3,000 recorded sessions, including 2,938 successful handshakes, ML-KEM-768 reduced mean handshake latency by 43% and connection energy by 34% relative to X25519, while the hybrid mode increased latency and energy by only 11% and 7%, respectively. However, full PQC exhibited a 3.8% connection-failure rate, likely caused by larger cryptographic payloads and buffer constraints. Applied to an industrial sensor profile, IoT-QARS identifies hybrid deployment as the most practical near-term migration option and highlights TCP-buffer tuning as a prerequisite for reliable full-PQC adoption. These results provide actionable guidance for post-quantum migration in embedded and industrial IoT environments.

Index Terms—embedded systems, ESP32, hybrid key exchange, Internet of Things, ML-KEM-768, post-quantum cryptography, TLS 1.3.

I. INTRODUCTION

The standardization of ML-KEM in NIST FIPS 203 marked a major step toward quantum-resistant cryptography [1], but practical deployment challenges remain, especially in the IoT. Many resource-constrained devices still depend on classical public-key mechanisms and are expected to remain in service for years, potentially into the era when quantum attacks become relevant [2], [15]. Thus, the key question is not whether migration is needed, but how urgently it must be carried out and at what cost.

In this context, the TLS 1.3 handshake is a critical point of analysis because it governs authentication and session key establishment, directly shaping computational, communication, and energy overhead [14], [16]. For constrained microcontrollers, these costs affect not only latency, but also memory

This research work was conducted as part of the execution of Project "Pionier-Q interconnection with the EuroQCI's space segment (24-EU-DIG-PIONIER-Q-SAT), Project no.: 101249721, funded by European Union under The Connecting Europe Facility (CEF) in Digital programme.

use, power consumption, and communication reliability, all of which are central to real-world deployment in battery-powered embedded systems.

The scientific problem addressed in this paper is the lack of experimentally grounded evidence on the performance and deployability of classical, full post-quantum, and hybrid TLS 1.3 key exchange on constrained Xtensa-class IoT hardware. Existing studies have investigated post-quantum cryptography on ARM-based embedded platforms, higher-capability edge devices, or server-class systems, and have shown that the main bottlenecks may shift between computation and communication depending on the platform and protocol design [3]–[5], [7]–[9]. Hybrid approaches have also been studied as a practical migration path, but their cost remains strongly dependent on the target hardware and network conditions [10]–[12]. Nevertheless, direct large-scale measurements on ESP32-class Xtensa hardware using wolfSSL over real TCP/IP communication remain limited. This gap makes it difficult for practitioners to judge whether post-quantum migration is technically feasible and operationally justified in widely deployed IoT device classes.

The aim of this work is to evaluate the authentication overhead of TLS 1.3 on the ESP32 platform under three key-exchange regimes: classical X25519, full post-quantum ML-KEM-768, and hybrid X25519+ML-KEM-768. The study seeks to determine how these alternatives affect handshake latency, end-to-end connection time, energy consumption, memory behavior, and connection reliability, and to convert these empirical findings into a practical migration-priority assessment for constrained IoT deployments.

The novelty of this paper is threefold. First, it provides, to the best of the authors' knowledge, the first direct large-scale comparison of classical, full-PQC, and hybrid TLS 1.3 key exchange on ESP32 Xtensa LX6 hardware using wolfSSL under identical experimental conditions. Second, it demonstrates that ML-KEM-768 can outperform classical X25519 in both handshake time and connection energy on this architecture, which is a counter-intuitive result for constrained embedded platforms. Third, it extends the PAREK Quantum-Adjusted Risk Score framework by introducing IoT-QARS, a device-oriented migration assessment model that incorporates empirically observed implementation overhead into migration-

priority scoring [6].

The main contributions of this paper are as follows:

- 1) an empirical dataset obtained from three 1000-session TLS 1.3 measurement campaigns on ESP32 using wolfSSL for classical, full-PQC, and hybrid key exchange;
- 2) a quantitative comparison of timing, energy, memory, and reliability characteristics across the three migration regimes;
- 3) the identification of a counter-intuitive performance result, namely that ML-KEM-768 is faster and more energy-efficient than X25519 on the evaluated Xtensa-class platform, while also revealing reliability limitations associated with larger cryptographic payloads;
- 4) the proposal of IoT-QARS as an extension of the PAREK framework for integrating measured implementation overhead into device-level post-quantum migration planning;

The remainder of this paper is organized as follows. Section II reviews related work on post-quantum cryptography in embedded systems, TLS overhead, and hybrid key exchange. Section III describes the experimental platform, configurations, and data-collection methodology. Section IV presents the empirical results. Section V introduces the IoT-QARS model and demonstrates its application to an industrial sensor profile. Section VI discusses the implications of the findings, including migration barriers, architectural observations, and deployment strategy. Finally, Section VII concludes the paper and outlines directions for future research.

II. RELATED WORKS

A. PQC on Embedded and IoT Platforms

Research on post-quantum cryptography for constrained platforms shows that deployment overhead depends strongly on device architecture, protocol design, and the trade-off between computation and communication. Bürstinghaus-Steinbach et al. [4] integrated Kyber and SPHINCS+ into mbed TLS and showed that Kyber-based key establishment can be competitive with classical elliptic-curve mechanisms, whereas large post-quantum signatures remain more challenging for deployment. Their study also demonstrated the feasibility of PQC experimentation on the ESP32-PICO. Similarly, Schöffel et al. [3] evaluated NIST Round 3 schemes on the nRF52840 and found that handshake latency is often driven more by bandwidth expansion than by computation, while battery impact may remain limited.

In broader IoT settings, Chung et al. [7] reported that several post-quantum candidates in MQTT-based environments can approach the overhead of classical asymmetric mechanisms, while Fitzgibbon and Ottaviani [8] identified CRYSTALS-Kyber and CRYSTALS-Dilithium as efficient choices for more capable IoT hardware such as Raspberry Pi 4. A closer methodological parallel to this work is provided by Tasopoulos et al. [17], [18], who evaluated PQC-enabled TLS 1.3 on resource-constrained systems using wolfSSL and showed that both performance and energy cost depend on the interaction

between algorithm choice and platform behavior. Together, these studies confirm the feasibility of post-quantum TLS on embedded devices, but they do not offer a unified large-scale ESP32 analysis directly comparing classical, full-PQC, and hybrid TLS 1.3 key exchange under a common measurement methodology.

B. PQC Overhead in TLS Handshakes and Authentication

At the protocol level, Sikeridis et al. [5] showed that post-quantum overhead in TLS 1.3 and SSH depends strongly on algorithm choice, transport configuration, and even buffer sizing, with modest TCP tuning able to reduce apparent slowdown. Alnahawi et al. [9] likewise conclude that Kyber-based TLS handshakes can in some cases approach or even outperform classical ECDHE. On the authentication side, Sikeridis et al. [19] demonstrated that certificate size, signature scheme, and certificate-chain structure significantly affect TLS 1.3 connection cost, while Paul et al. [20] explored mixed certificate chains as a practical migration path in wolfSSL. These studies help frame the present work: our experiments isolate ephemeral key-exchange overhead by retaining classical certificates, whereas full post-quantum deployment would also introduce authentication-side costs.

C. Hybrid Key Exchange and Alternative Migration Paths

Hybrid key exchange is widely viewed as a practical migration strategy because it retains classical security while adding protection against harvest-now-decrypt-later threats. Paquin et al. [10] showed that hybrid ECDHE+Kyber in TLS can remain close to classical performance under favorable conditions, although overhead becomes more visible under bandwidth constraints or packet loss. Related work extended this idea to triple-hybrid TLS 1.3 designs [11], while Ricci et al. [12] proposed a three-key combiner with formal security analysis and FPGA-oriented implementation considerations.

An alternative migration path is KEMTLS. Gonzalez and Wiggers [21] showed on embedded systems that protocol redesign can reduce handshake time, memory use, and traffic compared with conventional post-quantum TLS 1.3, indicating that migration cost depends not only on the cryptographic primitive but also on protocol architecture. From the standardization perspective, current IETF drafts now define both the general hybrid TLS 1.3 construction [22] and concrete ECDHE-MLKEM named groups such as X25519MLKEM768 and SecP256r1MLKEM768 [23], providing a formal basis for current implementations and experimental evaluations.

D. PQC Authentication Beyond TLS in IoT Protocols

Beyond TLS, Samandari and Gritti [13] evaluated post-quantum authentication in MQTT and showed that KEM-based authentication techniques can substantially reduce CPU-cycle cost compared with signature-based approaches. This result reinforces a broader observation across IoT-oriented post-quantum research: on constrained devices, the most practical migration path may depend as much on protocol design choices as on the raw performance of individual algorithms.

Post-quantum mechanisms are increasingly feasible on embedded platforms, but the reported overhead depends heavily on hardware architecture, network conditions, TLS stack, certificate strategy, and migration design. What remains insufficiently explored is a single-study comparison on ESP32-class Xtensa hardware that uses wolfSSL to evaluate classical, full-PQC, and hybrid TLS 1.3 key exchange under identical conditions, while also translating empirical performance and reliability measurements into a structured migration-priority model. This is the gap addressed by the present work.

III. EXPERIMENTAL METHODOLOGY

A. Hardware and Software Platform

All experiments were conducted on an ESP32 development board (NodeMCU ESP32 - Xtensa LX6 Dual-Core, up to 240 MHz, configured at 160MHz, 4 MB (configured 2MB) flash, 520 KB SRAM) running a custom TLS 1.3 client application built on wolfSSL 5.8.2. The device connected via Wi-Fi to a Raspberry Pi running an OpenSSL TLS server. Power was measured in-line using INA219 current monitor at approximately 500 Hz; bus voltage, mean current, mean power, and total energy were recorded per session alongside software-layer timing. Free heap was sampled before and after each handshake to quantify per-run allocation overhead; peak stack consumption per task was recorded across the experiment. All three configurations used classical RSA-2048 server certificates, isolating the ephemeral key-exchange overhead. The actual connection schema of an ESP32 (client) can be seen in Fig. 1.

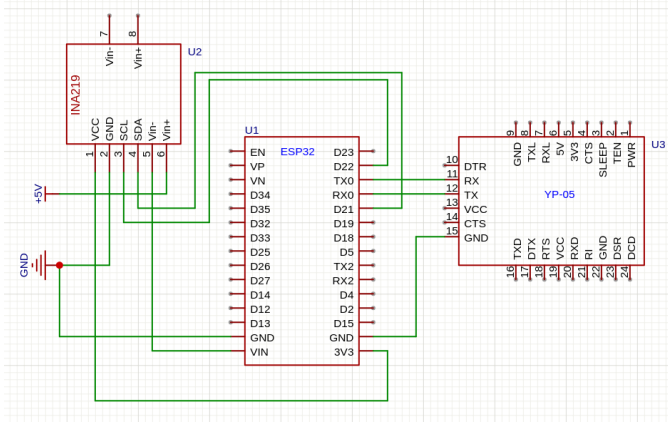


Fig. 1. ESP32 (client) wiring schematic

B. Experimental Configurations

Table I summarizes the three experimental configurations. ML-KEM-768 is standardized from CRYSTALS-Kyber768 under FIPS 203, providing NIST Security Level 3 (AES-192 equivalent classical security, 128-bit quantum security). X25519MLKEM768, defined in draft-kwiatkowski-tls-ecdhe-mlkem, combines classical and post-quantum key shares within the TLS 1.3 key schedule.

TABLE I
EXPERIMENTAL CONFIGURATIONS

ID	Mode	Key Exchange	Port	Attempted Sessions
C1	Classical	X25519	4433	1,000
C2	Full PQC	ML-KEM768	4434	1,000
C3	Hybrid	X25519+ML-KEM768	4435	1,000

C. Data Collection and Filtering

Each session record contains: config_id, mode, kex, status (ok/fail), total_us, tcp_connect_us, tls_handshake_us, heap_before/after, bus_voltage_V, avg_current_mA, avg_power_mW, energy_mJ, samples, sampling_span_us, among others. Statistical analysis was performed on status=ok sessions only. Failures in C2 and C3 manifest as wolfSSL error -313, consistent with TCP fragmentation of enlarged TLS records likely caused by ML-KEM-768 key material (1184-byte encapsulation key + 1088-byte ciphertext [1] vs. 32 bytes for X25519). The first session of each campaign was excluded from statistical analysis as a warmup run, as it reflected OS initialization overhead in addition to the handshake cost.

D. Testing Methodology Principles

With $n \geq 962$ successful runs per configuration, the standard error of the mean is well below 2 ms for all timing metrics. Energy per session was computed by integrating instantaneous power over time rather than using the mean-power approximation, making it a more informative figure of merit than mean power alone for duty-cycled IoT devices. Certificate type, server hardware, Wi-Fi channel, access point, physical separation between client and server, and wolfSSL build configuration were held constant across all three configurations; only the ephemeral key-exchange group differed.

IV. EXPERIMENTAL RESULTS

A. Dataset Overview

Table II summarizes session counts and failure rates. The zero failure rate for C1 confirms protocol and network stability. The elevated failure rates for C2 and C3 are consistent with record-layer buffer errors (wolfSSL error -313) likely caused by enlarged ML-KEM-768 key material.

TABLE II
SESSION COUNTS AND CONNECTION RELIABILITY

Config	Total	OK	Fail	Fail %
C1 - X25519	1,000	1,000	0	0.0 %
C2 - Full PQC	1,000	962	38	3.8 %
C3 - Hybrid	1,000	976	24	2.4 %

B. TLS Handshake Time

Table III presents the TLS handshake timing statistics. The most striking result is that C2 achieves a 43 % reduction in handshake time relative to C1. We attribute this to ML-KEM's Number Theoretic Transform (NTT) operations over small integer rings ($q = 3329$, 12-bit coefficients [1]) executing more efficiently on the Xtensa LX6 than X25519's 255-bit scalar multiplication, which requires extended-precision arithmetic chains absent in hardware. The coefficient of variation for C3 ($\sigma/\mu = 1.6\%$) is notably lower than for C1 (4.6 %), suggesting that the lattice component dominates and regularizes the timing profile. The hybrid C3 is 11 % slower than C1, reflecting the additive cost of executing both algorithms in sequence.

TABLE III
TLS HANDSHAKE TIME (MS) - SUCCESSFUL SESSIONS

Config	n	Mean	Std	Min	Max
C1 - X25519	999	911.6	41.6	863.4	1111.5
C2 - Full PQC	961	519.7	18.8	504.6	758.1
C3 - Hybrid	975	1011.3	16.2	977.2	1238.5

C. Total End-to-End Connection Time

Mean TCP connect time is comparable across configurations (C1: 20.7 ms, C2: 15.4 ms, C3: 14.1 ms), confirming that timing differences reflect TLS handshake computation rather than network-layer variation (Table IV).

TABLE IV
TOTAL END-TO-END CONNECTION TIME (MS)

Config	n	Mean	Std	Min	Max
C1 - X25519	999	941.4	56.4	881.9	1231.7
C2 - Full PQC	961	544.3	33.6	525.4	822.3
C3 - Hybrid	975	1034.5	28.3	997.0	1283.8

D. Energy Consumption

Table V shows the per-session energy figures. ML-KEM-768 (C2) consumes **34 % less energy** per TLS session than X25519 (C1). The hybrid (C3) consumes 7 % more. Although C2 has higher average power (476 mW vs. 420 mW for C1), its substantially shorter handshake duration yields lower total energy - a key insight for battery-powered duty-cycled devices.

TABLE V
ENERGY CONSUMPTION PER TLS SESSION (MJ)

Config	n	Mean	Std	Min	Max
C1 - X25519	999	395.2	47.9	322.3	602.0
C2 - Full PQC	961	261.4	28.6	232.7	435.2
C3 - Hybrid	975	424.0	36.8	382.1	624.2

E. Memory (Heap) Usage

Application-level heap consumption is comparable across all configurations (C1: 4018 B, C2: 4036 B, C3: 4025 B, $\sigma < 125$ B in all cases (Table VI)). The ML-KEM public key and ciphertext buffers are allocated internally by wolfSSL and do not appear in the application heap delta. The primary RAM risk lies in intermediate TLS record buffers, which are the likely source of the observed connection failures.

TABLE VI
HEAP CONSUMPTION PER TLS SESSION (BYTES)

Config	n	Mean	Std	Min	Max
C1 - X25519	999	4018	113	2220	4280
C2 - Full PQC	961	4036	112	3964	4280
C3 - Hybrid	975	4025	122	2184	4272

F. Overhead Summary

Table VII consolidates the overhead ratios. Handshake time results are broadly consistent with PQC survey findings [9] and the modest additive cost of hybrid combinations aligns with Paquin et al. [10]. The energy efficiency advantage of ML-KEM-768 on Xtensa LX6 is a new contribution not previously reported.

TABLE VII
OVERHEAD SUMMARY RELATIVE TO CLASSICAL BASELINE (C1)

Metric	C2/C1	C3/C1
TLS Handshake Time	0.57× (−43 %)	1.11× (+11 %)
Total Energy	0.66× (−34 %)	1.07× (+7 %)
Average Power	1.13× (+13 %)	0.97× (−3 %)
Heap Consumption	≈ 1.00×	≈ 1.00×
Failure Rate	3.8 % vs. 0 %	2.4 % vs. 0 %

V. IoT-QARS: PAREK-ALIGNED MIGRATION RISK MODEL

A. Motivation and Foundations

The PAREK framework [6] defines the Quantum-Adjusted Risk Score (QARS) as a five-factor linear model for cryptographic asset migration urgency:

$$\text{QARS} = w_1 \cdot \frac{T_{\text{shelf}}}{T_{\text{threat}}} + w_2 \cdot \frac{T_{\text{mig}}}{T_{\text{buf}}} + w_3 D_s + w_4 E_s + w_5 C_p \quad (1)$$

where each term is normalized to $[0, 1]$ (ratios capped at 2.0) and $\sum_i w_i = 1$.

The Mosca inequality [15] underpins w_1 : when the time needed to migrate (T_{mig}) plus the remaining confidentiality requirement (T_{shelf}) exceeds the expected CRQC arrival time (T_{threat}), immediate action is required. When applied to IoT device authentication, three additional dimensions become relevant beyond those of the standard organizational-level model:

- 1) *Operational overhead feasibility*: Is the PQC or hybrid scheme deployable within the energy budget, latency SLA, and reliability threshold of the target device class?

- 2) *Device lifetime vs. threat window*: Many IoT devices have 5-15-year field lifetimes that intersect the CRQC risk horizon; T_{shelf} must reflect *device deployment lifetime* rather than data confidentiality requirements.
- 3) *Connection reliability penalty*: Elevated handshake failure rates impose operational costs that must be weighed against security gains.

B. IoT-QARS Formula

We extend formula (1) with a sixth (w_6) factor:

$$\text{IoT-QARS} = w_1 \cdot \frac{T_{\text{dev}}}{T_{\text{threat}}} + w_2 \cdot \frac{T_{\text{mig}}}{T_{\text{buf}}} + w_3 D_s + w_4 E_s + w_5 C_p + w_6 \Phi_{\text{OH}} \quad (2)$$

where T_{dev} is the planned device field lifetime and $\Phi_{\text{OH}} \in [0, 1]$ is the *implementation feasibility penalty*:

$$\Phi_{\text{OH}} = \frac{1}{3} (\hat{\tau}_{\text{hs}} + \hat{E}_{\text{conn}} + \hat{F}_{\text{fail}}) \quad (3)$$

$\hat{\tau}_{\text{hs}}$, $\hat{E}_{\text{conn}}$, and $\hat{F}_{\text{fail}}$ are normalized (0 - 1) scores for handshake latency overhead, connection energy overhead, and failure rate respectively, computed against defined device-class budgets. The revised default weights for IoT contexts are shown in Table VIII.

TABLE VIII
IoT-QARS DEFAULT WEIGHTS

Weight	Factor	Default	Rationale
w_1	$T_{\text{dev}}/T_{\text{threat}}$	0.25	Device lifetime vs. CRQC
w_2	Migration difficulty	0.20	Firmware OTA complexity
w_3	Data sensitivity	0.20	Telemetry/control class.
w_4	Exposure surface	0.15	Network reachability
w_5	Compliance penalty	0.10	NIS2/sector obligations
w_6	Overhead feasibility	0.10	Deployability on target hardware

C. Normalized Overhead Scores for ESP32

Using device-class budgets of 1200 ms maximum TLS handshake, 600 mJ maximum per connection, and 2 % maximum acceptable failure rate, Table IX gives the normalized overhead scores. Full-PQC (C2) is computationally feasible but penalized by its failure rate; the hybrid (C3) carries both higher computation cost and the same failure penalty, motivating TCP-buffer tuning to improve deployment reliability.

TABLE IX
NORMALISED OVERHEAD SCORES FOR ESP32 CONFIGURATIONS

Config	$\hat{\tau}_{\text{hs}}$	$\hat{E}_{\text{conn}}$	$\hat{F}_{\text{fail}}$	Φ_{OH}
C1 - X25519	0.76	0.66	0.00	0.47
C2 - Full PQC	0.43	0.44	1.00	0.62
C3 - Hybrid	0.84	0.71	1.00	0.85

D. Full IoT-QARS Example: Industrial Sensor

Consider an industrial IoT sensor with $T_{\text{dev}} = 8$ yr, $T_{\text{threat}} = 10$ yr, $T_{\text{buf}} = 3$ yr, $T_{\text{mig}} = 2$ yr, $D_s = 0.70$ (critical infrastructure telemetry), $E_s = 0.50$, $C_p = 0.20$ (NIS2 essential entity). Table X presents the resulting IoT-QARS scores. The model correctly identifies the hybrid approach (C3) as the appropriate immediate migration target, providing Harvest-Now-Decrypt-Later (HNDL) protection while implementation overhead is known and bounded.

TABLE X
IoT-QARS FOR REPRESENTATIVE INDUSTRIAL SENSOR PROFILE

Scheme	QARS	IoT-QARS	Band
C1 - Retain classical	0.568	0.615	Medium
C2 - Full PQC only	0.568	0.630	Medium
C3 - Deploy hybrid	0.568	0.653	High

E. Integration with PAREK Layers

Following the PAREK three-layer aggregate model ($w_{\text{Transit}} = 0.40$, $w_{\text{Use}} = 0.20$, $w_{\text{Rest}} = 0.40$) [6], the Transit layer is dominant for devices conducting frequent cloud telemetry. In typical IoT deployments, Use and Rest scores are generally lower (session tokens are short-lived; small-deployment IoT archives are typically not encrypted at rest), increasing the relevance weight of our Transit-layer experimental findings.

VI. DISCUSSION

A. ML-KEM Computational Advantage on Xtensa LX6

The 43 % handshake speedup and 34 % energy reduction for ML-KEM arise from architectural factors. ML-KEM's NTT operations over small rings ($q = 3329$, 12-bit co-efficients [1]) map efficiently to Xtensa's 32-bit multiply-accumulate instructions. X25519, by contrast, requires 255-bit scalar multiplication demanding multi-word arithmetic chains that are inherently sequential on a processor without hardware big-integer support. This extends the conclusion of Schöffel et al. [3] - that computation is *not* the bottleneck for PQC in IoT TLS - by showing that on Xtensa, ML-KEM computation is an active *advantage* over classical ECC.

B. Failure Rate as the Principal Migration Barrier

The 3.8 % failure rate for C2 is the most significant operational obstacle to immediate full PQC deployment. Failures are likely triggered when the ML-KEM key share (2176 bytes total vs. 32 bytes for X25519) causes TLS record fragmentation exceeding wolfSSL's default receive buffer on ESP32. Three mitigation pathways are available: (1) increase the wolfSSL receive buffer size via appropriate build-time configuration; (2) implement application-layer retry with exponential back-off; (3) enable session resumption to amortize the one-time handshake cost. The hybrid C3 shows a lower failure rate (2.4 %).

C. Three-Phase Migration Strategy

Our results support a phased migration aligned with the PAREK R-stage [6]:

- **Phase 1 (Immediate):** Deploy hybrid (C3) with wolfSSL receive buffer enlargement and retry logic. Provides HNDL protection at 7 % energy overhead, relative to classical X25519 deployment.
- **Phase 2 (≤ 24 months):** Migrate to full PQC, realizing the 34 % energy saving and 43 % latency improvement after validating buffer-size fixes.
- **Phase 3 (post-CRQC):** Deprecate X25519 from hybrid (server-side ciphersuite pruning; no firmware update required).

D. Limitations

Results are specific to ESP32 Xtensa LX6; ARM Cortex-M4 and RISC-V platforms would yield different computational profiles. All configurations use classical certificates; adding ML-DSA (FIPS 204) server certificates would increase handshake size. Tests used a local Wi-Fi network with minimal packet loss; higher-latency or lossy networks would increase the relative impact of ML-KEM's larger ciphertext, per Sikeridis et al. [5]. The wolfSSL ML-KEM implementation may not be fully optimized for Xtensa; assembly-optimized NTT could further improve C2 and C3 performance.

VII. CONCLUSION

This paper presented the first empirical study of TLS 1.3 authentication overhead on Xtensa-class hardware, covering classical, full-PQC, and hybrid key exchange with $n \geq 962$ successful sessions per configuration. The principal finding - that ML-KEM-768 reduces TLS handshake time by 43 % and connection energy by 34 % relative to X25519, while the hybrid (C3) adds only 11 % latency and 7 % energy overhead - provides strong practical evidence supporting the immediate deployment of hybrid IoT firmware.

The proposed IoT-QARS extension of the PAREK framework translates these empirical findings into actionable migration priority scores. Applied to a representative industrial sensor profile, it classifies hybrid deployment as High priority and identifies TCP-buffer tuning as the prerequisite for full-PQC rollout.

Future work will incorporate ML-DSA server certificates, evaluate the RISC-V ESP32-C3 variant, and extend the IoT-QARS model to include supply-chain attack surfaces relevant to NIS2 critical infrastructure operators.

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